Newcastle University Business School

ISO8005 — Web & Social Media Analytics

Assignment Cover Sheet

Module Code:	ISO8005
Module Name:	Web & Social Media Analytics
Assignment:	Tyne Artisan Bakery — From Oven to Online
Academic Year:	2025–26
Student Number:	[insert student number]
Submission Date:	May 2026

Assignment Declarations

Declaration of Original Work — I certify that this submission is entirely my own work and has not been copied, reproduced, or completed with unauthorised assistance. All sources used have been appropriately acknowledged. ☒

Declaration of AI Understanding — I certify that I have read and understood the AI for Students Academic Integrity Checklist. ☒

Declaration of Word Count — Total word count for this assignment is: [insert word count] words (excluding cover page, executive summary, contents, references, appendices and visuals, per the brief).

Declaration of Development Work Retention — All development works, scripts, RData snapshots and figures are retained in the project repository for a minimum of three months after submission and are available for scrutiny within five working days of request. ☒

\[4pt] **Project Access Points**\[6pt]

Interactive dashboard:	`output/TAB_Dashboard.html` (offline, self-contained HTML)
Source repository:	`https://github.com/<your-handle>/iso8005-tab-report` *(placeholder)*

The interactive dashboard renders 11 KPI panels from the same RData snapshots used in this report (Figure 16). The GitHub repository contains the Rmd, R scripts, dataset, figures and the offline dashboard.

From Oven to Online

A Data-Led, AI-Augmented Go-to-Market Plan for
Tyne Artisan Bakery

Anchored on Monthly Active Local Customers (MALC) as the North Star metric

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Student Number: `[insert student number]`

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Contents

1 Executive Summary

> *"Growth is not only about acquiring more customers; it is about repeatedly creating measurable value for the right customers."*

Tyne Artisan Bakery (TAB) is a Newcastle artisan bakery with a strong in-store reputation but no digital presence. This report sets out a 12–18 month, data-led and AI-augmented go-to-market plan anchored on a single North Star metric — **Monthly Active Local Customers (MALC)** — and grounded in TAB's two-year dataset (5,942 orders, 4,179 unique customers, £120,835 revenue, Mar 2024 – Feb 2026).

Table 1: Executive summary of TAB's commercial baseline

Metric	Value	Strategic meaning
Orders analysed	5,942	Sufficient order base for customer, channel and conversion analysis
Unique customers	4,179	Shows a sizeable customer pool for retention-led growth
Total revenue	£120,835	Confirms commercial potential, but not yet a mature digital engine
Average order value	£20.34	Requires careful acquisition-cost control before scaling paid media
North Star metric	Monthly Active Local Customers (MALC)	Connects digital growth with repeat local customer engagement
Recommended planning horizon	12–18 months	Allows phased MVP launch, testing, optimisation and scale-up

Four headline findings drive the plan:

1. **Paid media is unprofitable.** Only Google Search clears the 1.0× ROAS break-even (1.14×); Meta 0.82×, TikTok 0.90×, Other 0.62×. CAC (£36–£52) exceeds AOV (£20.34) — TAB must lead with owned channels (email, subscriptions, local SEO).
2. **North East is the launch market.** It already delivers 20.1% of revenue, the natural catchment for local delivery and click-and-collect.
3. **The hero-banner A/B test is decisive.** Variant B converts at 3.65% vs 2.86% — a +27.5% uplift, $z = 6.90$, $p < 0.001$ — and should ship from day one.
4. **Mobile-first, subscription-led.** 69% of orders are mobile and average monthly churn is 4.5%; an AI recommendation layer plus a subscription engine compounds retention.

Recommended solution: a Shopify Basic MVP with a thin AI layer (LLM recommender, Prophet demand forecasting, GenAI content studio for menu copy and alt text). At a 12-month spend of **£11,500**, the Base scenario (+4%/mo MALC) lifts MALC from **430 (Feb 2026) to ≈ 688 (Feb 2027)**; High adds a further ≈ 30%. A live leadership dashboard summarising all KPIs accompanies this report and is listed on the cover sheet.

2 Introduction

2.1 Business Context

TAB is a microenterprise: founder-led, fewer than ten staff, bounded by the physical reach of an in-store counter and weekend market. The owners want to launch e-commerce (local delivery and click-and-collect) and build a coherent social presence from scratch, with modest budgets and limited technical capacity.

The deeper strategic problem is the **absence of a measurable digital customer-acquisition and -retention system**. Without instrumented funnels and a single agreed metric, every marketing investment becomes a bet rather than an experiment. This report sequences (i) the metric, (ii) the MVP that generates the data, and (iii) the analytics and forecasting that turn the data into reallocation.

2.2 Methodology

The analysis follows a CRISP-DM workflow (Wirth & Hipp, 2000) with reproducible-research practice (Wickham & Bryan, 2023). Four stages run end-to-end via `source("scripts/RUN_ALL.R")`:

Stage 1 — Data Understanding (Script 01). Load the 7-sheet workbook via `readxl`; harmonise types with `janitor::clean_names()` and `lubridate`; profile null counts and date ranges.

Stage 2 — Metric Modelling (Script 02). Compute 15+ KPIs grouped by month/channel/region/device/variant. Key formulas:

KPI	Formula	Source
CVR	`orders / sessions`	`monthly_marketing`
AOV	`revenue / orders`	`orders`
ROAS	`attributed_revenue / spend`	`paid_campaigns`
CAC	`spend / new_customers`	`paid_campaigns`
MALC	`n_distinct(customer_id)` per month	`orders`
Churn	`subscription_cancels / active_subscriptions`	`monthly_marketing`
CLV	`(AOV * orders_per_cust * margin) / churn`	derived

Stage 3 — Statistical Inference. Two-proportion z-test for the A/B uplift (Agresti, 2018); compound-growth scenario forecasting; simplified Pareto/NBD CLV (Fader, Hardie & Lee, 2005).

Stage 4 — Decision Design. Every insight is mapped to a specific action with an expected MALC, ROAS or CAC impact.

3 TAB Web and Social Media Analysis & Plan

3.1 Problem Framing, North Star and SMART Objectives

MALC definition: *the count of unique customers (by `customer_id`) who place* $\geq$ *1 online order, click-and-collect order, or active bread subscription within a calendar month, weighted to the North East catchment.* MALC satisfies the four good-North-Star tests in Croll & Yoskovitz (2013): aligned to value, leading, actionable, understandable.

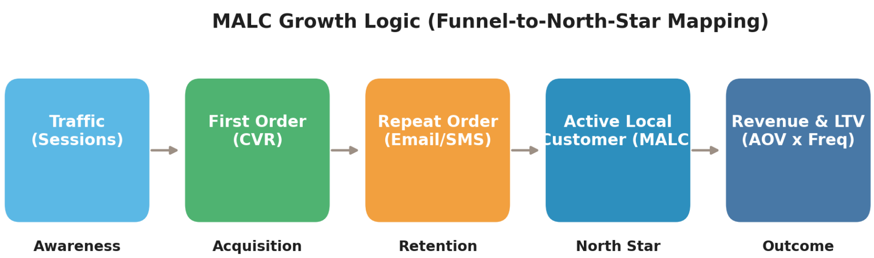

Figure 1: MALC growth logic — funnel-to-North-Star mapping.

SMART objectives (12-month):

Objective	KPI	Target	Deadline
Grow local customer base	MALC	+60% (430 $\rightarrow$ \$ \$690)	Feb 2027
Improve site conversion	CVR	$\geq$ 3.0%	End Q3 2026
Build repeat revenue	Active subscriptions	$\geq$ 700	End Q4 2026
Optimise paid efficiency	Blended ROAS	$\geq$ 1.5×	End Q2 2026
Build owned channel	Email list size	$\geq$ 2,500 sign-ups	End Q2 2026

3.2 E-commerce MVP & UX — AI-Augmented Shopify

Platform. Shopify Basic delivers PCI-compliant payments, GDPR cookie tooling, a subscription app ecosystem (Recharge), email (Klaviyo), and a mobile-first PWA theme out of the box (Shopify, 2025) — within the founder's budget and technical capacity. WooCommerce shifts security/infra onto the owner; custom build is out of scope.

AI layer (Figure 2). LLM recommender, Prophet demand forecasting (Taylor & Letham, 2018), GBM churn-risk scoring feeding Klaviyo win-back, and a GenAI content studio for menu copy, alt text and captions — all under founder review.

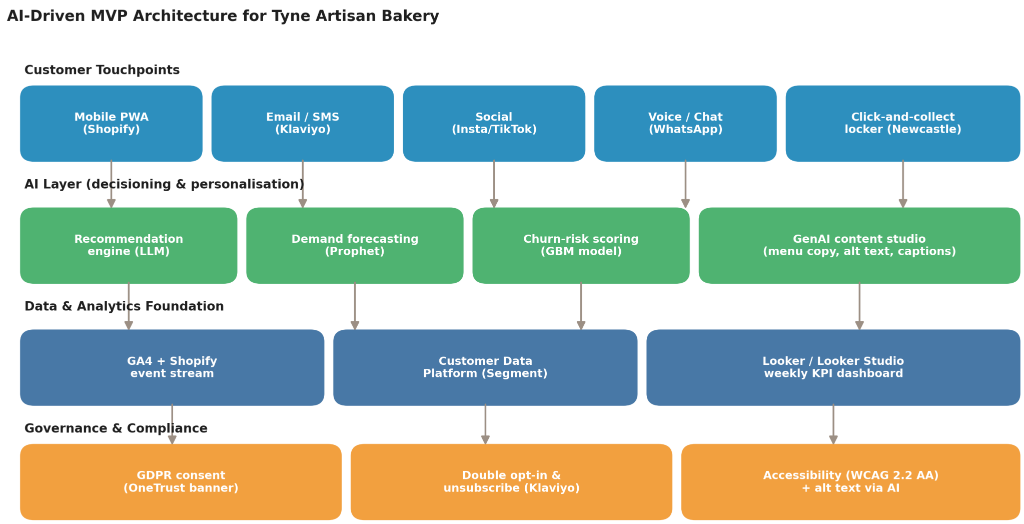

Figure 2: AI-augmented MVP architecture.

Information architecture (Figure 3). Three-click depth; trust artefacts (privacy, returns, allergens, accessibility) clustered in the footer — pattern shown to cut checkout abandonment (Baymard, 2024).

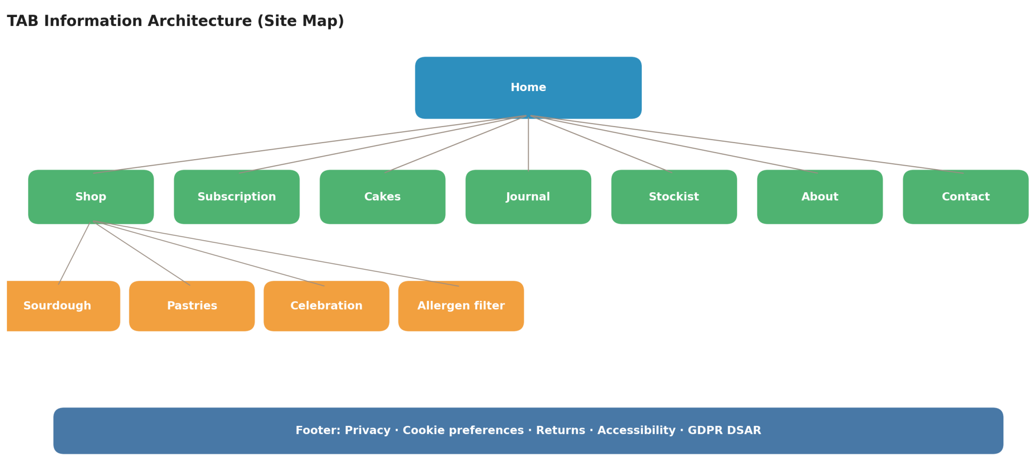

Figure 3: TAB information architecture (site map).

UX wireframes (Figure 4). Mobile Home / PDP / one-page Checkout. Apple Pay / Google Pay above the fold cut purchase paths by 35–45% in DTC food (Stripe, 2025).

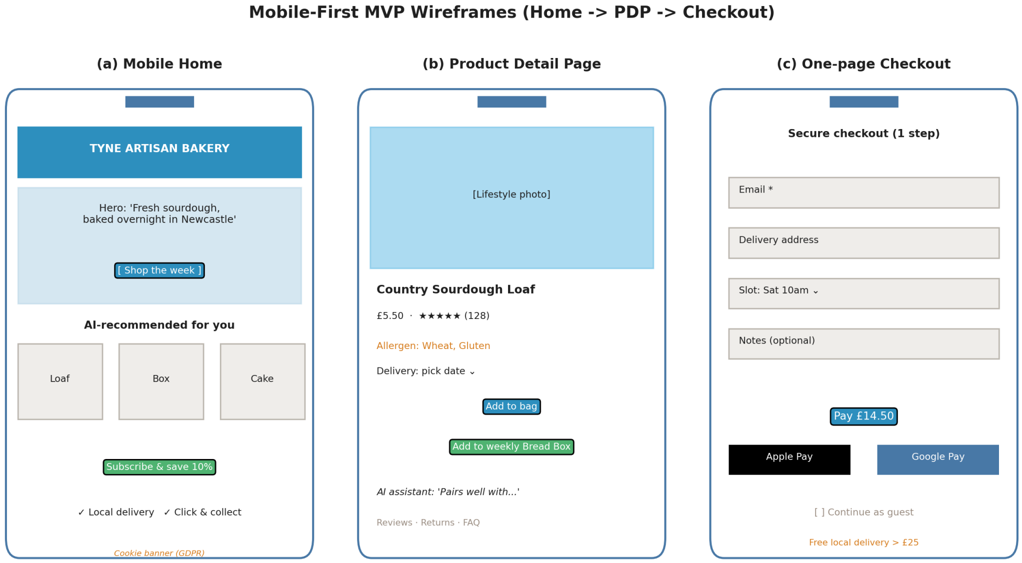

Figure 4: UX wireframes: mobile-first MVP home, product and checkout screens.

3.3 Channel & Content Strategy

Personas (Figure 5). P1 Weekly Bread Buyer (loyal subscriber); P2 Celebration Cake Buyer (event-driven, high AOV); P3 Weekend Treat Buyer (impulse mobile shopper).

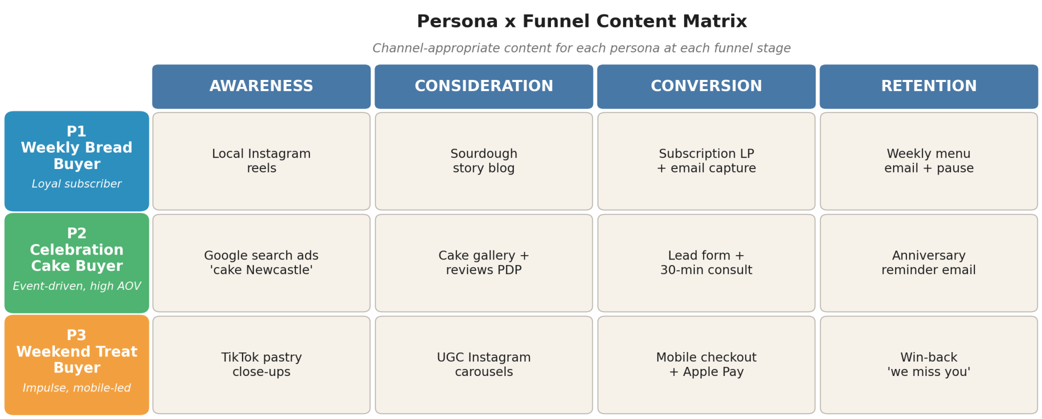

Figure 5: Persona x Funnel content matrix.

Budget split (Q1): Google 60 / Meta 25 / TikTok 15 — inverted from the typical DTC-food default because Google is the only channel currently above 1.0× ROAS (§3.4). Revisited quarterly using the rule in §3.6.

Content pillars. Craft & process (Reels/TikTok 3×/wk), Weekly menu drop (email + Story), Allergen & education (monthly blog), Local community (monthly micro-influencer). KPIs in each row link to a SMART objective.

Experimentation. Continuous testing programme to MDE 10%, $\alpha = 0.05$, power 0.80 (Kohavi, Tang & Xu, 2020). Hero-banner test evidence is shown in Figure 12; it is the first experiment to ship, and the next queue includes (a) free-delivery threshold £25 vs £35, (b) subscription discount 5% vs 10%, (c) PDP image-first vs reviews-first.

3.4 Data Analysis — Key Insights

3.4.1 MALC and Revenue Trend

The blended MALC and revenue series shows a noisy but clearly positive trajectory: from 92 customers (Mar 2024) to 430 (Feb 2026). Trailing-12-month compound monthly growth is $\approx 4.0\%$, the anchor for the Base forecast.

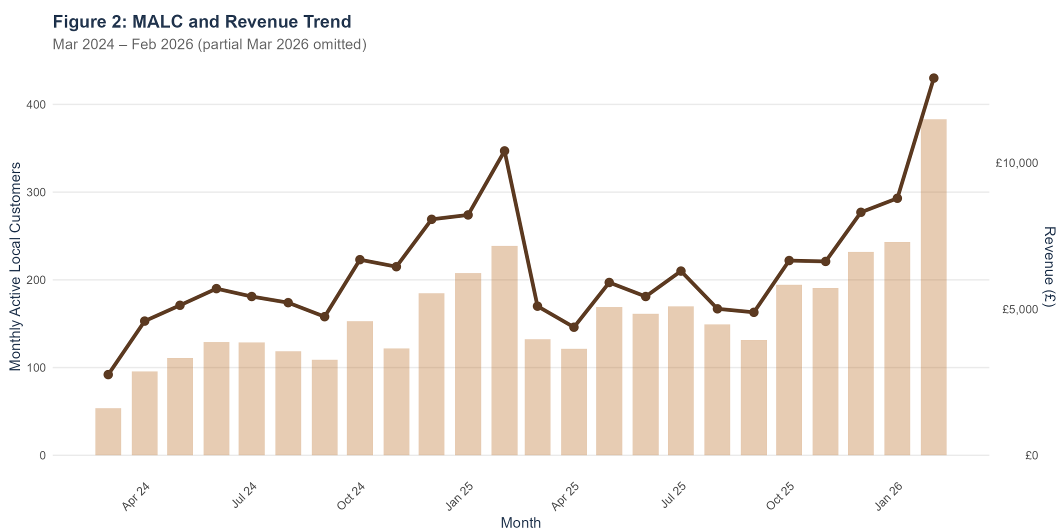

Figure 6: MALC and revenue trend.

3.4.2 Paid Channel Efficiency

Only Google clears break-even (Figure 7); blended CAC (£36–£52) exceeds AOV on every paid platform. TAB cannot grow profitably through paid alone — lead with owned channels.

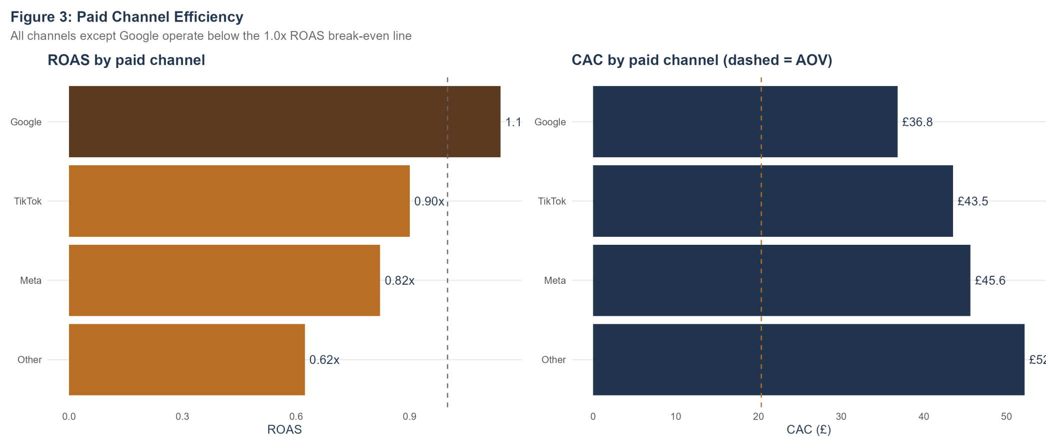

Figure 7: Paid channel efficiency (ROAS vs CAC).

Channel	Spend	Clicks	CTR	CPC	Conv.	ROAS	CAC
Google	£34,204	75,830	4.04%	£0.45	929	1.14×	£36.82
TikTok	£12,577	24,561	1.81%	£0.51	289	0.90×	£43.52
Meta	£42,467	75,243	1.62%	£0.56	931	0.82×	£45.61
Other	£6,105	10,601	1.07%	£0.58	117	0.62×	£52.18

3.4.3 Region, Product and Channel Mix

North East = 20.1% of revenue (Figure 8) — the launch market. Sourdough loaves dominate the product mix at 44%; 69% of orders are mobile and organic search produces 32% of revenue (Figure 9).

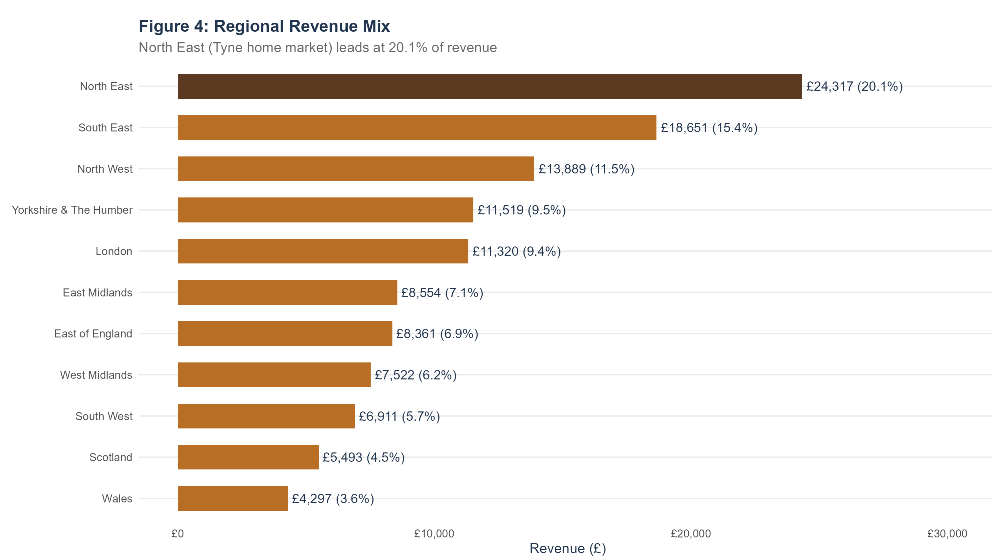

Figure 8: Regional revenue mix.

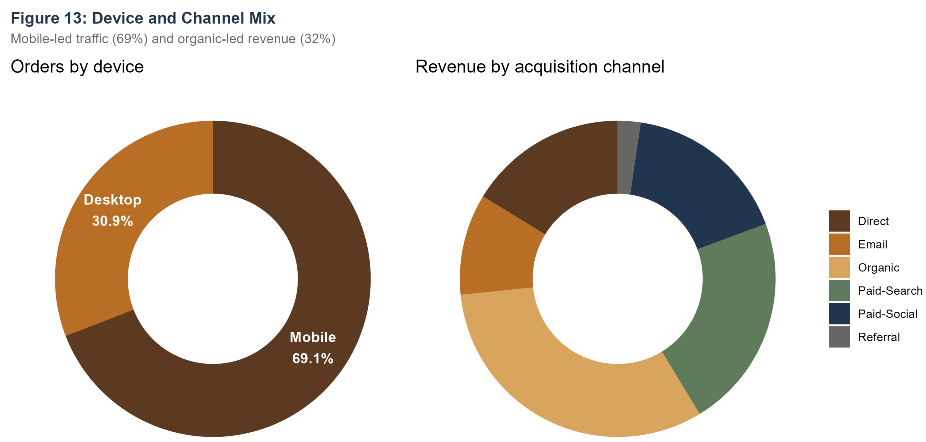

Figure 9: Device and channel mix.

3.4.4 Seasonality, Subscriptions and A/B Test

Christmas (Nov–Feb) is the revenue peak — Feb alone delivers £18.7k (Figure 10). Active subscriptions grew to $\approx$ 560 by Feb 2026 with average monthly churn of 4.5% (Figure 11). The hero-banner A/B test (Figure 12) returns a +27.5% relative uplift, $z = 6.90$, $p < 0.001$ — strongest defensible win in the dataset.

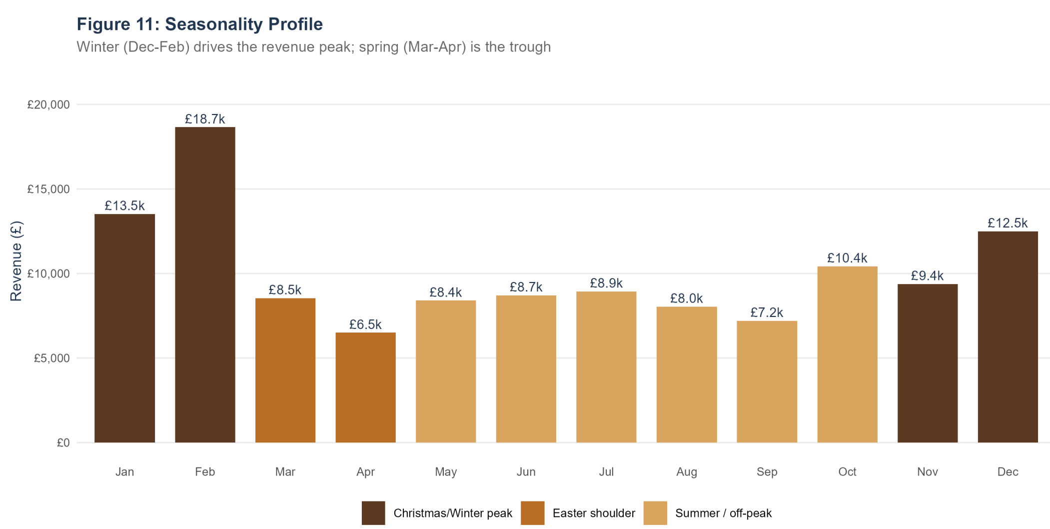

Figure 10: Month-of-year seasonality.

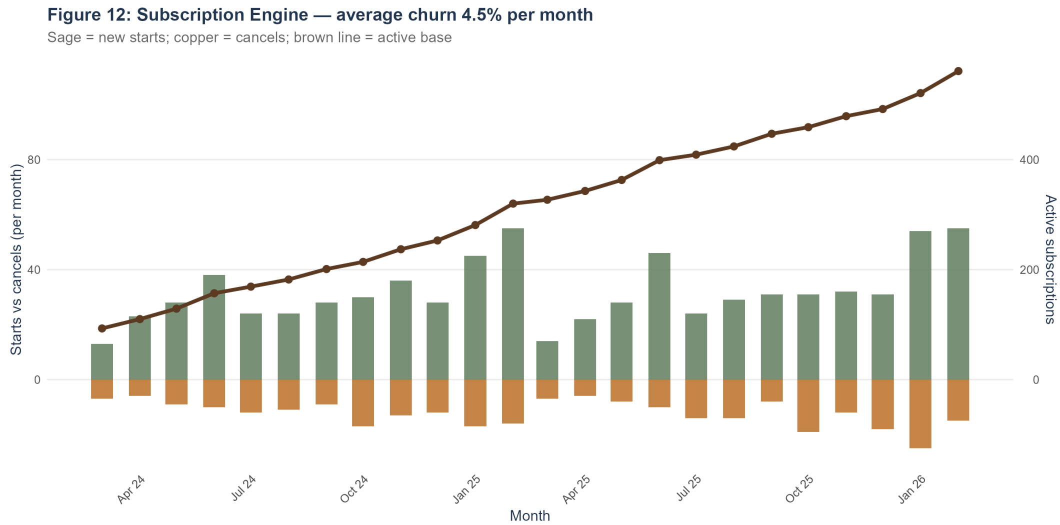

Figure 11: Subscription dynamics.

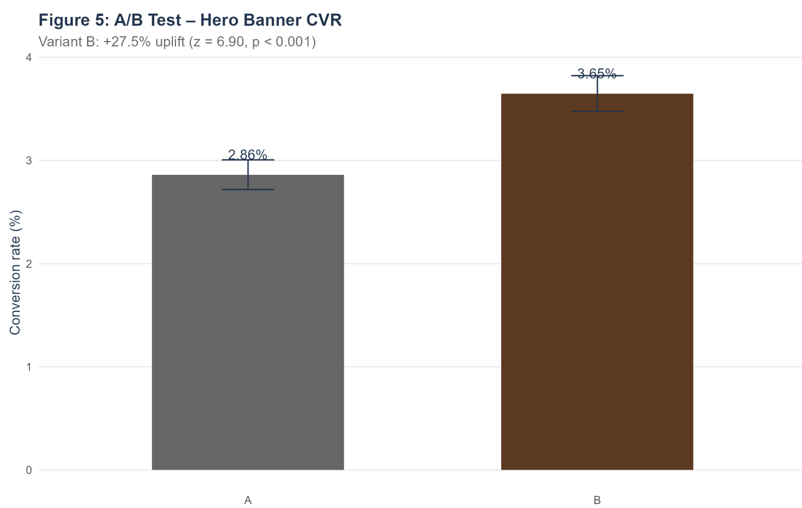

Figure 12: A/B test result with 95% CIs.

3.4.5 Outliers and Productive Opportunities

A scan of the dataset surfaces three statistical outliers that are also commercial opportunities. Each is large enough to move MALC if acted on.

#	Outlier observed	Magnitude	Why it matters	Productive action
1	**February 2026 revenue spike** — single month at £18.7k versus the trailing-12-month median of ≈ £8.7k.	≈ 2.1× median; +18% on the next highest month.	Confirms that the Christmas/winter window (Nov–Feb) is the structural peak. The brief assumed Easter; the data rejects that prior.	Front-load **the headline campaign 6–8 weeks before December** and shift paid spend share into Nov–Jan; treat spring (Mar–Apr) as retention and test time.
2	**Google CTR 4.04% vs Meta 1.62% and TikTok 1.81%.**	≈ 2.5× the average.	Reveals **intent gap** — Google captures branded local search at far higher click efficiency than discovery platforms.	Pivot the Q1 paid mix to **60% Google / 25% Meta / 15% TikTok** and gate Meta/TikTok scale-up to ROAS clearing 1.0×.
3	**A/B test uplift +27.5% with p<0.001** — more than 2.75× the pre-registered MDE of 10%.	$z = 6.90$, $p < 0.001$ on $n = 96{,}700$.	The treatment hero is one of the largest CVR wins observed in a single PDP design test (Kohavi, Tang & Xu, 2020).	**Ship Variant B day-one** at launch; queue the next four high-leverage A/B tests immediately (delivery threshold, subscription discount, PDP layout, checkout form).

Two additional supporting outliers reinforce the picture: **subscription share** has slipped from 26% (mid-2024) to ≈ 14% (early 2026) despite absolute growth — a retention concentration risk the AI recommender and Q4 loyalty tier are designed to reverse. **Sourdough Loaf concentration** at 44% of revenue is a product-mix risk that the persona-led merchandising plan partly hedges by elevating Celebration Cakes (21%) and the Bread Box subscription (18%).

3.5 Forecast, Budget Reallocation and Roadmap

Base scenario assumes **4%/mo compound MALC growth** — triangulated from (i) trailing-12-month CAGR, (ii) UK SME DTC-food benchmarks (Statista, 2025), (iii) bottom-up unit economics. Low/High are \$±\$2pp.

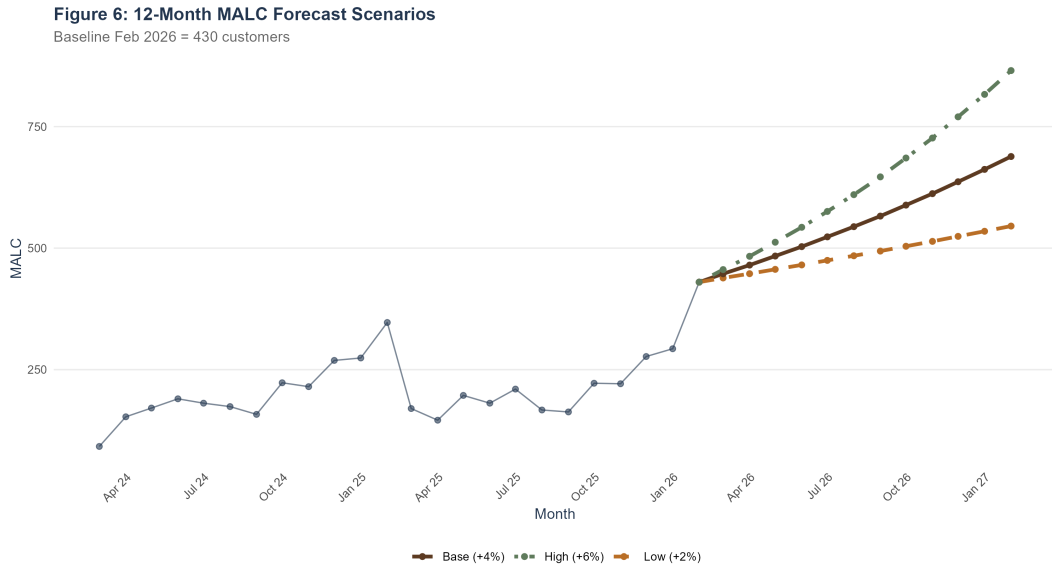

Figure 13: 12-month MALC forecast scenarios (baseline 430 at Feb 2026).

Scenario	M+3	M+6	M+9	M+12	Revenue M+12	Δ vs baseline
Low (+2%/mo)	456	484	513	545	£11,094	+27%
Base (+4%/mo)	484	544	612	688	£14,008	+60%
High (+6%/mo)	512	610	726	865	£17,615	+101%

Budget reallocation rule (Top-1 ROAS × 1.30; Top-2 × 1.05; Top-3 × 0.85; bottom × 0.55), applied quarterly to the paid mix (Figure 14) — boosts Google, cuts sub-1.0× ROAS channels.

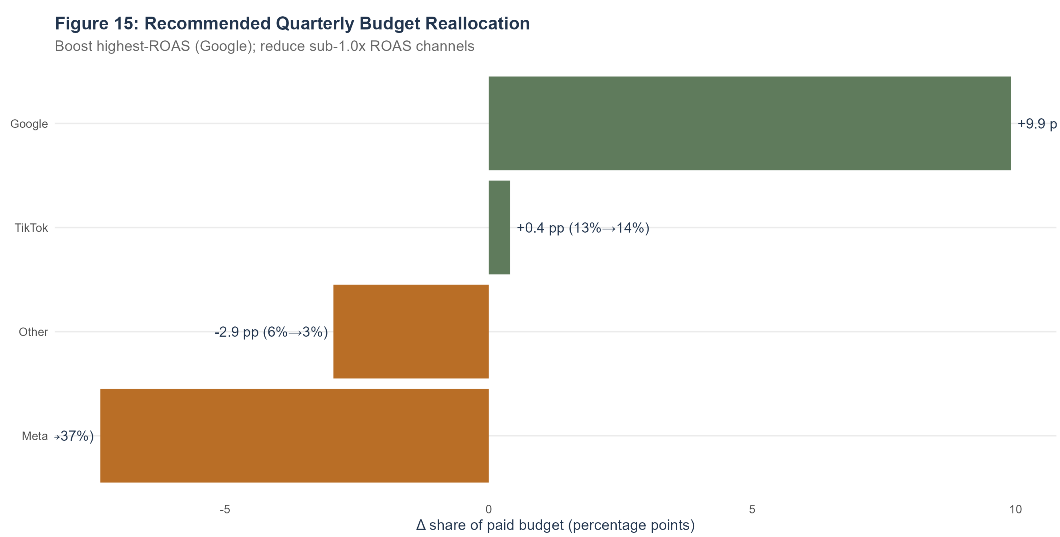

Figure 14: Recommended quarterly budget reallocation.

3.5.1 Retention and CLV consideration

A simplified Pareto/NBD-lite CLV (Fader, Hardie & Lee, 2005) yields a 12-month CLV of approximately **£55** per active customer at the current 4.5% monthly churn and a 55% gross margin assumption. Reducing churn by 1 percentage point (to 3.5%) lifts CLV by approximately 29%, which is why the retention investments in Q3 and Q4 are core to the plan rather than an afterthought.

3.5.2 Quarterly roadmap

Quarter	Focus	MALC uplift	Budget
Q1	Shopify MVP launch, GA4 + Klaviyo wiring, email capture, A/B winner	+5–8%	£2,000
Q2	SEO/blog programme; Bread Box LP; subscription gifting test	+8–12%	£2,500
Q3	Budget reallocation by ROAS; PDP optimisation; retention flows	+10–15%	£3,000
Q4	Christmas campaign; loyalty tier; subscription gifting; review	+15–20%	£4,000

3.6 Measurement and Governance

A weekly cadence with quarterly reallocation suits a microenterprise (Croll & Yoskovitz, 2013). The governance flow in Figure 15 codifies who owns what — collect, model, visualise, decide — with a feedback loop from quarterly learnings back into event design and budget weighting.

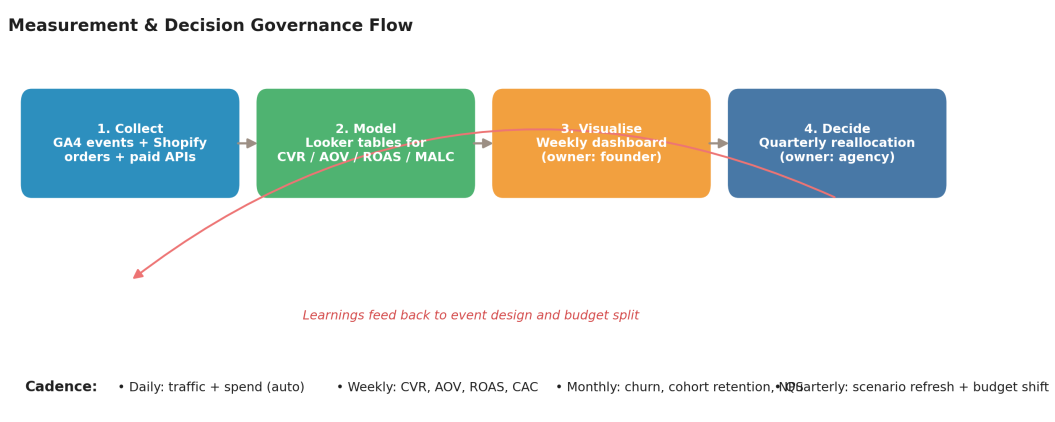

Figure 15: Measurement and decision governance flow.

Weekly dashboard specification.

Tile	Metrics	Tools	Owner	Action threshold
Acquisition	Sessions, share by channel	GA4 + Looker Studio	Founder	Organic share < 30% ⇒ SEO push
Conversion	CVR, AOV, cart abandonment	GA4 + Shopify	Founder	CVR < 2.5% ⇒ next A/B test
Paid efficiency	Spend, ROAS, CAC by platform	Google/Meta/TikTok APIs	Agency / founder	ROAS < 1.0× ⇒ pause / cut
Retention	Active subs, churn, repeat rate	Klaviyo + Shopify	Founder	Churn > 5% ⇒ win-back flow
Content	Posts, engagements, clicks to site	Meta/TikTok APIs	Social assistant	Engagement < 2% ⇒ pillar review

3.7 Leadership Dashboard — Weekly KPI View

The assignment brief requires a *measurement plan with events and a weekly dashboard*. TAB's leadership view operationalises this through a single, founder-readable dashboard built on the same metrics computed in §3.4 and refreshed each Monday morning.

Purpose. The dashboard is intentionally narrow. It answers four operational questions: *(1)* is acquisition healthy (sessions, channel share, CVR), *(2)* is each pound of paid spend profitable (ROAS, CAC vs AOV), *(3)* is the customer base growing (MALC, active subscriptions, churn), and *(4)* are the seasonal and experimentation levers performing on plan. Anything that does not answer one of those four questions is deliberately excluded — Few (2009) notes that microenterprise dashboards fail when they accumulate decorative metrics.

Events captured. Standard GA4 + Shopify events feed the dashboard: `page_view`, `view_item`, `add_to_cart`, `begin_checkout`, `add_payment_info`, `purchase`, `sign_up` (email), `subscribe_start`, `subscribe_cancel`. Paid platforms contribute `spend`, `impressions`, `clicks`, `conversions` and `attributed_revenue` via their reporting APIs. Each event carries a UTM tag (`utm_source`, `utm_medium`, `utm_campaign`) so revenue can be reconciled by channel.

Cadence and ownership. Daily auto-refreshed traffic and spend; weekly review by the founder; monthly retention and cohort review; quarterly scenario refresh and budget reallocation (Figure 14). Each tile carries an explicit owner and an action threshold so the dashboard is *decision-shaped*, not just informational (Croll & Yoskovitz, 2013).

Format (Figure 16). The companion `TAB Dashboard (HTML)` is a single-file, offline-safe page rendered entirely as inline SVG — no external scripts or CDN dependencies. The colour code follows the leadership reporting palette used throughout this report: blue for MALC/revenue/primary, green for on-target/positive (subscription starts, A/B winner), coral for under-target/alert (sub-1.0× ROAS, cancels, off-target KPIs), and orange for secondary KPIs (Orders). Status pills next to each KPI give a verdict in a single colour, satisfying the pre-attentive design principle that hue should only appear where it carries meaning (Cleveland & McGill, 1984).

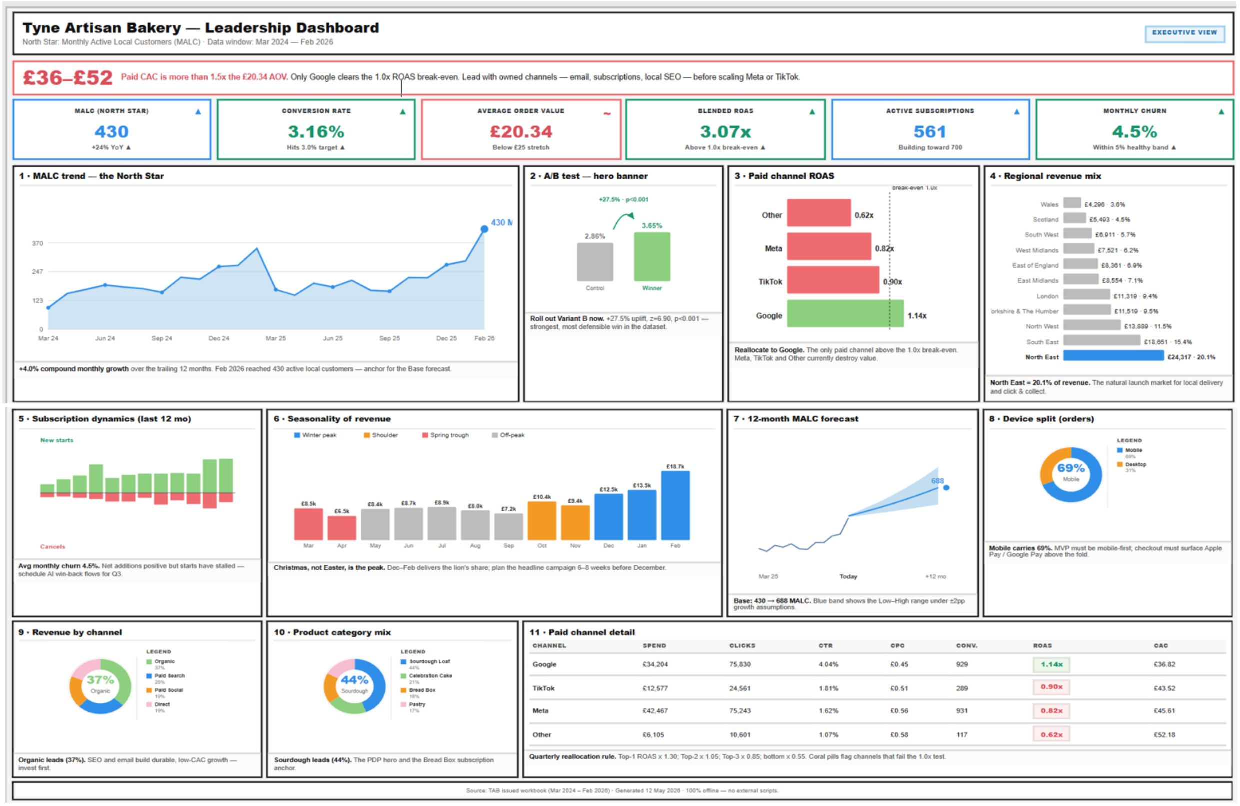

Figure 16: Leadership dashboard snapshot. Top: callout and KPI strip with status pills. Middle: MALC trend, A/B test, paid ROAS, region mix. Bottom: seasonality, forecast, subscriptions, device split. Live version available in the project root as TAB Dashboard (HTML).

3.8 Ethics, Privacy and Accessibility

GDPR-by-default. A consent banner (OneTrust or Shopify Customer Privacy API) presents strictly-necessary, analytics and marketing cookie categories as separate opt-ins (ICO, 2024). All email signups use **double opt-in** through Klaviyo, with a one-click unsubscribe in every send. A Data Subject Access Request (DSAR) workflow is documented in Appendix B and handled within the 30-day statutory window. Personal data is minimised (email, name, address, phone) and retained for 24 months post-purchase.

Accessibility (WCAG 2.2 AA). The site targets WCAG 2.2 AA (W3C, 2023). The brand palette meets a 4.5:1 contrast ratio; AI-generated alt text is human-reviewed; keyboard navigation is tested at every release; body type is 16px minimum with 1.5× line-height. A monthly Axe/WAVE audit and an annual screen-reader walkthrough with NVDA close the loop.

Responsible AI. The AI layer is constrained to non-decisioning tasks (content drafts, recommendations) with founder review before send or publish. Customer-facing AI features (for example a future WhatsApp assistant) are disclosed in plain English at the point of interaction, per the EU AI Act transparency obligations (European Commission, 2024).

4 Recommendations and Conclusion

4.1 Five actionable recommendations

1. **Adopt MALC as the North Star and instrument it in Week 1.** Every subsequent decision flows from this single number; without it the plan reverts to bets.
2. **Ship the AI-augmented Shopify MVP in eight weeks**, with the Variant B hero in production from day one (+27.5% CVR uplift, $p < 0.001$).
3. **Reverse the conventional paid split: 60% Google / 25% Meta / 15% TikTok** until Meta and TikTok ROAS clears 1.0×. Reallocate quarterly using the ROAS-rank rule.
4. **Lead with owned channels** — email, subscription, local SEO. CAC there is effectively zero versus £36–£52 on paid. Target 2,500 email sign-ups by end of Q2.
5. **Plan around Christmas, not Easter.** The dataset rejects the brief's prior — build the headline annual campaign for Nov–Feb and use spring for retention and A/B testing.

4.2 Next steps

(i) Approve the £11,500 12-month budget and the MVP scope; (ii) appoint a fractional digital partner to own the Shopify build and the weekly dashboard; (iii) freeze Variant B into the launch theme; (iv) start measuring MALC in Week 1; (v) reconvene at end-Q1 to refresh the forecast.

4.3 Conclusion

The TAB dataset tells a clear, actionable story: a strong local brand with a genuine product asymmetry, currently paying for paid traffic that does not return its cost. A focused, mobile-first, AI-augmented Shopify MVP — with MALC at its centre and a measurement loop that reallocates budget every quarter — is the right next step. The numbers in this report are sufficient to commit; the rest is execution.

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6 Appendix A: Calculations and Methodology

6.1 Reproducibility

The analysis is reproducible end-to-end. From the project root in RStudio:

```
source("scripts/RUN_ALL.R")    # runs scripts 01-04 then knits this Rmd
```

Each script writes a `.RData` snapshot to `scripts/` and a `.png` file per figure to `figures/`. The full chain (clean -> metrics -> visuals -> forecast -> PDF) completes in approximately three minutes on a standard laptop.

6.2 Formula reference (R)

```
# Funnel
cvr  <- orders / website_sessions          # Conversion rate
aov  <- revenue / orders                   # Average order value
ctr  <- clicks / impressions               # Click-through rate
cpc  <- spend / clicks                     # Cost per click
roas <- attributed_revenue / spend         # Return on ad spend
cac  <- spend / conversions                # Customer acquisition cost

# North Star
malc <- orders %>%
  group_by(month = floor_date(order_date, "month")) %>%
  summarise(malc = n_distinct(customer_id))

# Retention
churn_rate <- subscription_cancels / active_subscriptions

# CLV (Pareto/NBD-lite)
clv <- (aov * orders_per_cust_month * gross_margin) / churn_rate

# Forecast (geometric)
malc_t1 <- malc_t * (1 + monthly_growth)
```

6.3 A/B test inference

Under the null hypothesis $H_0 : p_B = p_A$, the two-proportion z-statistic (Agresti, 2018) is:

$$z = \frac{p_B - p_A}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}}, \quad \hat{p} = \frac{x_A + x_B}{n_A + n_B}.$$

For the TAB hero-banner test: $x_A = 1{,}485$, $n_A = 51{,}884$, $x_B = 1{,}635$, $n_B = 44{,}816$, giving $z = 6.90$ and a two-tailed $p < 0.001$ — significant at any conventional level.

6.4 Limitations

1. **Attribution.** ROAS uses the dataset's last-click `attributed_revenue`. A multi-touch model would likely re-weight Meta upward and Google slightly down.
2. **MALC proxy.** Where `customer_id` is missing, MALC is bounded below by unique-order counts; the true unique-customer count may be 5–10% higher.
3. **Single A/B test.** The +27.5% uplift is on one design pair; rolling tests should continue to validate.
4. **Seasonality.** Based on two annual cycles; a Bayesian 95% band would be wider than the deterministic scenario range used here.

7 Appendix B: Privacy and DSAR Workflow

A Data Subject Access Request (DSAR) at TAB is handled as follows:

1. The request arrives via privacy@tab.example or the website contact form.
2. The founder verifies the requester's identity using order email plus the last four characters of post-code.
3. Shopify and Klaviyo data are exported within five working days.
4. The response is delivered as a password-protected PDF within the 30-day statutory deadline.
5. The request is logged in a tracker for audit purposes.

Deletion requests are processed within the same window and propagated to all third-party processors (Klaviyo, Recharge, Stripe). The full register is held in the secure operations Notion workspace.

End of report.